

EDITORIAL



CAR-T cells for H3K27-altered diffuse midline gliomas: where do we stand?

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Among the pediatric brain tumors, diffuse midline gliomas (DMG) harboring a mutation in histone H3K27 have the deadliest prognosis [1]. These tumors are most commonly found in the midline structures of the brain resulting in cranial nerve palsies and diplopia at clinical presentation [1]. Standard of care is radiation therapy although this is predominately palliative. At a molecular level, lysine (K) 27 is most commonly replaced by a methionine (M) but regardless of the specific histone H3 mutation, there is a global reduction in H3K27 trimethylation. This mutation most often occurs at one of two genes; most commonly the H3F3A gene resulting in the canonical H3.3K27M but also the HIST1H3B resulting in H3.1K27M [1–3]. Although these tumors have a relatively low mutational burden in comparison to other cancers, other mutations contributing to tumorigenesis have been documented including TP53 and PDGFRA, both of which are implicated in as driving mutations of proliferation in a variety of cancers [4,5]. Over the decades, conventional chemotherapy and more recently, targeted agents have still failed to provide any survival benefit [6].

With other cancers yielding to new immune-based therapies, many have sought to implement similar strategies into the treatment of DMG. There are multiple ongoing or recently completed clinical trials evaluating the use of checkpoint inhibitors although a study with PD1 administration did not show any benefit compared with radiation therapy alone [7]. Vaccine-based therapies are also under preclinical and clinical evaluation. Recent results from early clinical trials demonstrated the safety and clinical effect in some patients following administration of a H3.3K27M-specific peptide vaccine with phase 2 studies now underway, including one in combination with a PD1 checkpoint inhibitor [1,7]. Other peptide-based vaccines targeting EGFRvIII and surviving in addition to dendritic cell vaccines are in clinical trial as

well [1,7]. The final immunotherapy under investigation is CAR-T cell therapy, which will be discussed in further detail below.

Immunotherapies are dependent on the tumor immune microenvironment. With increased tissue samples being acquired via surgical biopsies, studies have further elucidated details pertaining to the DMG immune microenvironment [8]. There is sparse infiltration of CD3⁺ and CD8⁺ lymphocytes as well as few NK cells although, in comparison to pediatric high grade gliomas (pHGG), there were more CD4⁺ Treg cells, eosinophils, neutrophils and dendritic cells. Additionally, there are very low levels or no proinflammatory cytokines and limited chemokines [7,8]. For example, there are significantly lower levels of TGFβ when compared with hemispheric pHGG [7]. These findings have resulted in experts labeling K27-altered DMG as immunologically “cold”. There is some evidence to suggest a difference between H3.1-K27 altered and H3.3K27-altered tumors, with the former expressing a more immunosuppressive state and the latter having a more inert immune status [7]. The roles of adjuvant therapies can also modulate the immune microenvironment. Corticosteroid use can further suppress the global immune environment, including that of the tumor while radiation therapy may impact the tumor immune microenvironment through multiple mechanisms that can increase suppression while increasing antigen presentation [8]. The exact biological mechanism and the consequences of RT to the immune microenvironment in these tumors is an area of current study [8]. As more information pertaining to the DMG tumor microenvironment is uncovered, immune-related therapies will need to be adjusted accordingly to maximize their potential for success.

Chimeric antigen receptor (CAR)-T cells are autologously-derived engineered receptors that bind to a tumor specific antigen and activate cytotoxic

lymphocytes, usually T cells, that can subsequently elicit strong antitumor responses [9]. CARs are composed of four components: an extracellular antigen-binding domain, a hinge region, a transmembrane domain, an intracellular signaling domain [8,9]. The antigen-binding domain needs to be specific to the tumor cells and allow for binding of the CAR with high affinity allowing for the transduction of signals responsible for activating T cells [8]. The hinge region, which can be altered for optimal efficacy, serves to overcome any steric hindrance and optimize binding capability [9]. The transmembrane domain serves multiple functions, including anchoring the receptor into the T cell membrane and contributing to CAR stability and expression [9]. Finally, the intracellular signaling domain, which is classically derived from the CD3 ζ tyrosine kinase signaling motif, allow for complete activation [9]. Like all cancer therapies, CAR-T cells are not immune to mechanisms of escape or to off-target effects, however they have shown significant promise, particularly in hematologic malignancies leading to the first CAR-T therapy to be approved by the FDA for the treatment of B cell malignancies in 2017 [9]. There are many clinical trials currently investigating CAR-T therapy for both hematological and solid tumors, including DMG [8,9].

CAR-T cell success is largely dependent on the selection of a tumor specific antigen, which needs to confer strong affinity and adhesion for optimal CAR binding but is also homogeneously expressed on tumor cells and absent from healthy cells [10]. When bound to the tumor specific antigen, the effector T cell can drive expression of proinflammatory cytokines and release perforin and granzyme B, effectively destroying tumor cells [10,11]. However, on target but off-tumor effects and local and systemic toxicities occur when the chosen antigen is present on other cells throughout the body. In addition to selecting a tumor specific antigen, additional modifications to CAR design may result in increased improved adhesion, tracking, anti-tumor activity and/or reduce associated toxicities [11]. An example of this is the addition of cytokine/chemokine receptor genes [12]. With these considerations, there are a number of different CAR-T cell designs currently under clinical investigation for DMG.

The initial preclinical and clinical studies for CAR-T therapy targeted the B7-H3 (CD276) antigen, which is over-expressed in DIPG/DMG along with many other cancers including glioblastoma, medulloblastoma and meningioma [13]. Preliminary results from the ongoing trial (NCT04185038) were published in January of 2023. They engineered a B7-H3 targeted CAR that was transduced into equal populations of CD4⁺ and CD8⁺ T cells before intraventricular infusion back into the patients [14]. In

their initial cohort, the authors demonstrated the safety and feasibility of repetitive infusions of B7-H3 CAR-T cells although a larger patient cohort and long-term follow-up is needed [14]. Multiple studies demonstrated that H3K27M cells highly express disialoganglioside GD2 and given its low expression in healthy tissue, GD2 has become a popular antigen for CAR targeting [8,15]. Early results of four patients using a GD2 CAR-T cell with a 4-1BB co-stimulatory domain and a CD3 ζ signaling domain were published in 2022 [16]. This dose-escalation phase 1 study demonstrated clinical and radiographic improvement in 75% of patients and while significant toxicities were experienced by all patients, extensive supportive care resulted in successful management [16]. There are currently two active GD2 CAR-T cell clinical trials actively recruiting (NCT04196413, NCT04099797). The most recently initiated clinical trial (NCT05768880) is that of a world first quad targeting autologous CAR-T cell therapy where CARs engineered to express a combination of B7-H3, EGFR806, HER2 and IL-13 zetakine are delivered intraventricularly. No results are yet available although this a novel and promising approach.

There are a couple other trials in addition to those aforementioned, including those targeting EGFR, HER2 and IL13R 2 [8]. In all the trials, there seems to be two primary delivery techniques, intravenously (IV) or intraventricularly (ICV). One of the advantages of ICV delivery may be to avoid some of the more systemic toxicities associated with IV delivery. There also may be a greater potential to elicit the desired localized inflammatory reaction with ICV delivery although drastic inflammatory responses within the CNS, particularly the brainstem, may prove to be life threatening [13]. A further consideration for both the IV and ICV is the propensity for the CAR-T cells and other immune components to penetrate the blood-brain barrier (BBB) or blood CSF barrier (BCB) to reach the tumor cells. The brain is generally considered an immune-privileged organ with a limited number of microglia normally found so it is conceivable that T-cell would struggle to overcome the natural barriers designed to protect the brain. Current studies discussed above have shown evidence for detection of their CAR-T cells in the CSF and serum of patients although their CNS penetration remains largely unknown [14,16]. At autopsy of one patient in the GD2 CAR-T cell trial demonstrated detectable CAR-T cells in the CNS although there would be no way to confirm penetrance prior to significant progression and death [16].

There is much excitement surrounding CAR-T cell therapy for DMG especially with preliminary results showing promising clinical results as well as paving the path to further understand the implications for immunotherapies in CNS malignancies. With a demonstrated safety profile

for systemic and locoregional infusions, efficacy could be enhanced with more studies aimed at understanding CAR-T cell distribution and CNS penetration. It may be feasible to include a fluorescent tag on cells or similar strategy to allow for real-time distribution following infusion. An attempt at an intratumoral infusion may be considered, using a strategy such as convection-enhanced delivery to directly infuse the CAR-T cells into the tumor to bypass the BBB although the benefits of this must be weighed against the risk of an insurmountable inflammatory reaction or other negative delivery-related effects. With all things considered, the continued efforts of experts around the world working to develop novel CAR-T cell and other therapies continue to offer hope for the future for the children afflicted with this deadly disease.

Author contributions

Conception and design: EA Power, JS Rechberger, DJ Daniels. Drafting the article: EA Power, E Millesi, JS Rechberger. Critically revising the article: DJ Daniels. Reviewed submitted version of manuscript: EA Power, E Millesi, JS Rechberger, DJ Daniels. Approved the final version of the manuscript on behalf of all authors: JS Rechberger. Article supervision: DJ Daniels.

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●● This source is of considerable interest because it discusses some of the primary limitations thus far in the clinical experience for those using CART cells for the treatment of DMG.